

Network Monitoring, NetFlow & Packet Analysis

Three lenses

Flow (NetFlow/sFlow): src,dst,port,bytes,pkts – cheap, aggregates

Trace (PCAP/Zeek): full content/payload – deep, noisy

Structured (Zeek/Bro): per-protocol conn logs – the sweet spot

Collecting

- NetFlow on the router/FW → collector (nfdump, ElasticFlow, PRTG)
- Zeek on a mirror port → conn, http, dns, ssl logs
- PCAP archive for deep dive (have-a-day retention, not unlimited)

Detection patterns in the wire

Pattern	Signature	Appeared in
Beaconing	periodic same-size POST to same dst	C2
DGA	random-looking subdomains	botnets
Data exfil	sustained high bytes from a host post-login	theft
Port scan	many dst ports from one src in short time	recon
Zero-day C2	new external IP + same-timestamp conns	novel tools

The netmon playbook

- Baseline: traffic profiles per host/app (bytes/s, connections)
- Watch anomaly: high flow count, new egress destinations
- Verify with Zeek: what protocol actually ran?
- Drill with tshark/PCAP when needed

Hands-on

```
tshark -r big.pcap -q -z io,stat,10 # traffic overview
tshark -r big.pcap -Y 'dns' -T fields -e dns.qry.name | head
nfdump -r nfcapd.xxx 'host 10.0.0.5' | head
```

Blindspots

Encrypted payload (TLS 1.3 = opaque) – rely on metadata + JA3 (fingerprint) + behavior

NetFlow can't see content; PCAP it all for storage is rare

Flow tells you 'where'; Zeek tells you 'what protocol'; PCAP answers 'what happened'. Together they make traffic legible.

[← Segmentation, VPN & ZTNA](#)

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